Practice Problem 2.15

For the bridge network in Fig. 2.54, find R_{ab} and i.

Answer: 60 Ω, 4 A.

$\Delta \rightarrow Y$: $\boxed{\dfrac{\text{相邻之积}}{\text{周长之和}}}$ 或 $Y \rightarrow \Delta$: $\boxed{\dfrac{\text{两两相乘之和}}{\text{对面电阻}}}$

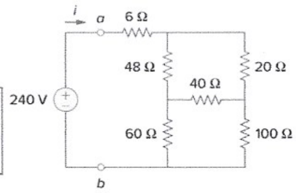
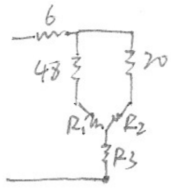


Figure 2.54
For Practice Prob. 2.15.

$\Rightarrow$



把右下角的40.60.100 ▽ 化成 Y

$$R_1 = \frac{2400}{200} = 12\ ;\ R_2 = \frac{4000}{200} = 20\ ;\ R_3 = \frac{6000}{200} = 30$$

$$\therefore R_{ab} = 6 + (48+12)//(20+20) + 30 = 36 + 60//40 = 36 + 24 = 60\,\Omega$$

$$i = \frac{240}{60} = 4A$$

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Practice Problem 3.2

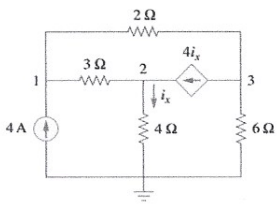

Figure 3.6
For Practice Prob. 3.2.

Find the voltages at the three nonreference nodes in the circuit of Fig. 3.6.

Answer: $v_1 = 32$ V, $v_2 = -25.6$ V, $v_3 = 62.4$ V.

节点电压法，②含受控电流源情况

node 1: $4 = \dfrac{V_1 - V_3}{2} + \dfrac{V_1 - V_2}{3}$

node 2: $\dfrac{V_1 - V_2}{3} + 4\cdot\dfrac{V_2}{4} = \dfrac{V_2}{4}$

node 3: $\dfrac{V_1 - V_3}{2} = 4\cdot\dfrac{V_2}{4} + \dfrac{V_3}{6}$

$$\Rightarrow \begin{cases} 5V_1 - 2V_2 - 3V_3 = 24 \\ 4V_1 + 5V_2 = 0 \\ 3V_1 - 6V_2 - 4V_3 = 0 \end{cases} \Rightarrow \begin{bmatrix} 5 & -2 & -3 \\ 4 & 5 & 0 \\ 3 & -6 & -4 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 24 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 32 \\ -25.6 \\ 62.4 \end{bmatrix} V$$

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Practice Problem 3.4

Find v_1, v_2, and v_3 in the circuit of Fig. 3.14 using nodal analysis.

Answer: $v_1 = 7.608$ V, $v_2 = -17.39$ V, $v_3 = 1.6305$ V.

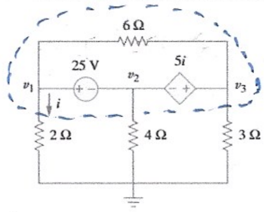

Figure 3.14
For Practice Prob. 3.4.

节点电压法，④含受控电压源情况

把虚线框当作广义结点：

$$\begin{cases} \frac{V_1}{2} + \frac{V_2}{4} + \frac{V_3}{3} = 0 \\ V_1 - V_2 = 25 \\ -V_2 + V_3 = 5 \cdot \frac{V_1}{2} \end{cases}$$

$$\Rightarrow \begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{3} \\ 1 & -1 & 0 \\ -\frac{5}{2} & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 25 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 7.609 \\ -17.391 \\ 1.63 \end{bmatrix} V$$

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Practice Problem 3.6

Using mesh analysis, find I_o in the circuit of Fig. 3.21.

Answer: -4 A.

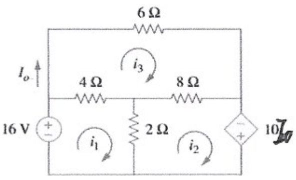

Figure 3.21
For Practice Prob. 3.6.

网孔电流法，②含受控电压源情况

loop 1: $-16 + (i_1 - i_3)\cdot 4 + (i_1 - i_2)\,2 = 0$

loop 2: $(i_2 - i_1)\cdot 2 + (i_2 - i_3)\cdot 8 - 10\,i_3 = 0$

loop 3: $(i_3 - i_1)\cdot 4 + i_3 \cdot 6 + (i_3 - i_2)\cdot 8 = 0$

$$\Rightarrow \begin{matrix} 6i_1 - 2i_2 - 4i_3 = 16 \\ -2i_1 + 10i_2 - 18i_3 = 0 \\ -4i_1 - 8i_2 + 18i_3 = 0 \end{matrix} \Rightarrow \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} -2.571 \\ -7.714 \\ -4 \end{bmatrix} A$$

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Practice Problem 3.7

Use mesh analysis to determine i_1, i_2, and i_3 in Fig. 3.25.

$i_1 = 4.8\ \text{A} \qquad i_2 = 0.8\ \text{A} \qquad i_3 = 1.6\ \text{A}$

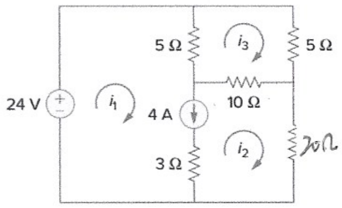

Figure 3.25
For Practice Prob. 3.7.

书上答案有错

$\boxed{\text{网孔电流法，③含电流源情况}}$

去掉4A branch，把loop1和loop2看成广义网孔

$-24 + (i_1 - i_3)\cdot 5 + (i_2 - i_3)\cdot 10 + i_2 \cdot 20 = 0$

loop3：

$(i_3 - i_1)\cdot 5 + i_3 \cdot 5 + (i_3 - i_2)\cdot 10 = 0$

另：

$4 = i_1 - i_2$

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$$\Rightarrow \begin{aligned} 5i_1 + 30i_2 - 15i_3 &= 24 \\ -5i_1 - 10i_2 + 20i_3 &= 0 \\ i_1 - i_2 &= 4 \end{aligned}$$

$$\Rightarrow \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} 4.8 \\ 0.8 \\ 1.6 \end{bmatrix} \text{A}$$